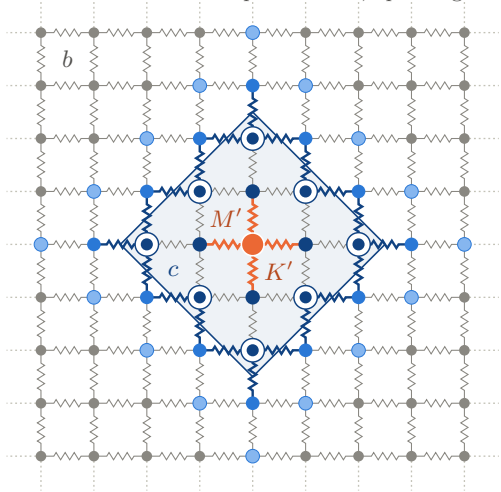


(a) Lattice, defect and cluster

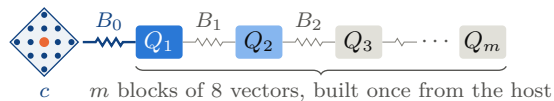
centre of the 121×121 square lattice, open edges



- defect, mass M'
- defect springs K'
- cluster c : $|i| + |j| \leq 2$ (13)
- ⊙ boundary of c (8 sites)
- $\mathcal{D}_{bc}$: 20 springs, rank 8
- Q_1 : shell $|i| + |j| = 3$
- Q_2 : shell $|i| + |j| = 4$
- bath b : $M = 1$, $K = 1$

(b) Upfolded matrix $\tilde{C}$

the bath b becomes a block Lanczos chain



$\tilde{C} =$

$$\begin{bmatrix} \boxed{\mathcal{D}_{cc}} & B_0^\dagger & & & \\ B_0 & A_1 & B_1^\dagger & & \\ & B_1 & A_2 & B_2^\dagger & \\ & & B_2 & A_3 & \\ & & & \ddots & \\ & & & & A_m \end{bmatrix}$$

 changes with the defect

all other blocks: host only,
one chain for the pristine
lattice and both defects

one diagonalization of $\tilde{C}$
gives $D_{cc}(\omega)$ at every ω

$\dim \tilde{C} = 13 + 8m$
($m = 160: 1293$)

$$D_{cc}(\omega) = \left[\left((\omega + i\eta)^2 - \tilde{C} \right)^{-1} \right]_{cc}$$

defect	M'	K'	signature ($\text{host band } 0 \leq \omega \leq 2\sqrt{2}$)
light	0.3	1.3	local mode at $\omega = 4.349$, above the band
heavy	8	0.6	resonance inside the band near $\omega = 0.45$